

Customer Data Cleaning and Revenue Analysis (excel)

Executive Summary	1
Data Cleaning Summary	1
Dataset Overview	1
Key Findings (with screenshots)	1
Age Distribution	2
Spending Trends	2
Screenshots (Graphs)	2
1. Total Customer Spend by Country	3
2. Monthly New Customer Sign-Ups	4
3. Customer Distribution by Country	5
Recommendations.....	5
Conclusion	6

Executive Summary

This project focused on improving the quality of a customer dataset and generating business insights through data analysis in Microsoft Excel. The dataset contained 838 customer records with issues such as missing age values, inconsistent date formats, non-standardized country names, and special characters within text fields.

Following the data cleaning process, the dataset was analyzed to identify customer demographics, spending behavior, and geographic trends.

Data Cleaning Summary

The following data quality issues were addressed:

Replaced missing age values using median imputation.

Standardized multiple date formats into a consistent structure.

Cleaned customer notes by removing special characters.

Standardized country names and corrected inconsistencies.

Validated numeric fields for analysis readiness.

These steps improved data accuracy, consistency, and usability for reporting purposes.

Dataset Overview

- Total Records: 838
- Countries Represented: 15
- Average Customer Age: 42.5 years
- Median Customer Age: 43 years
- Average Customer Spend: 347.17

Key Findings (with screenshots)

Customer Spending by Country

The highest total customer spending was recorded in:

- United States – 50,454.41
- United Kingdom – 41,446.38
- UAE – 22,685.98
- China – 21,161.27
- Egypt – 18,275.61

These markets contribute the largest share of customer revenue and may represent key opportunities for retention and growth initiatives.

Age Distribution

The largest customer segment falls within the 36–45 age group, accounting for approximately 45% of the dataset.

Age Group Breakdown:

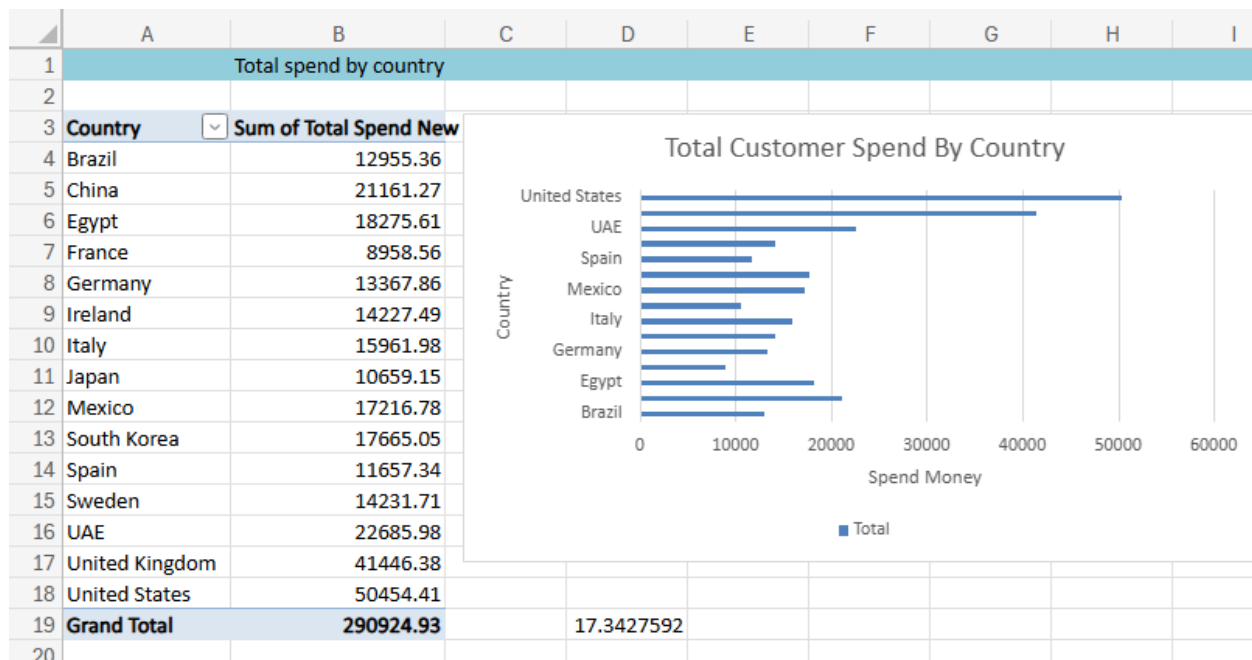
- 18–25: 82 customers
- 26–35: 122 customers
- 36–45: 375 customers
- 46–55: 130 customers
- 56–65: 129 customers

This indicates that middle-aged customers form the core customer base.

Spending Trends

Customer spending varies significantly across the dataset, ranging from 0 to 1,491.56. A large portion of customers recorded little or no spending activity, suggesting opportunities for targeted engagement and conversion campaigns.

Screenshots (Graphs)



1. Total Customer Spend by Country

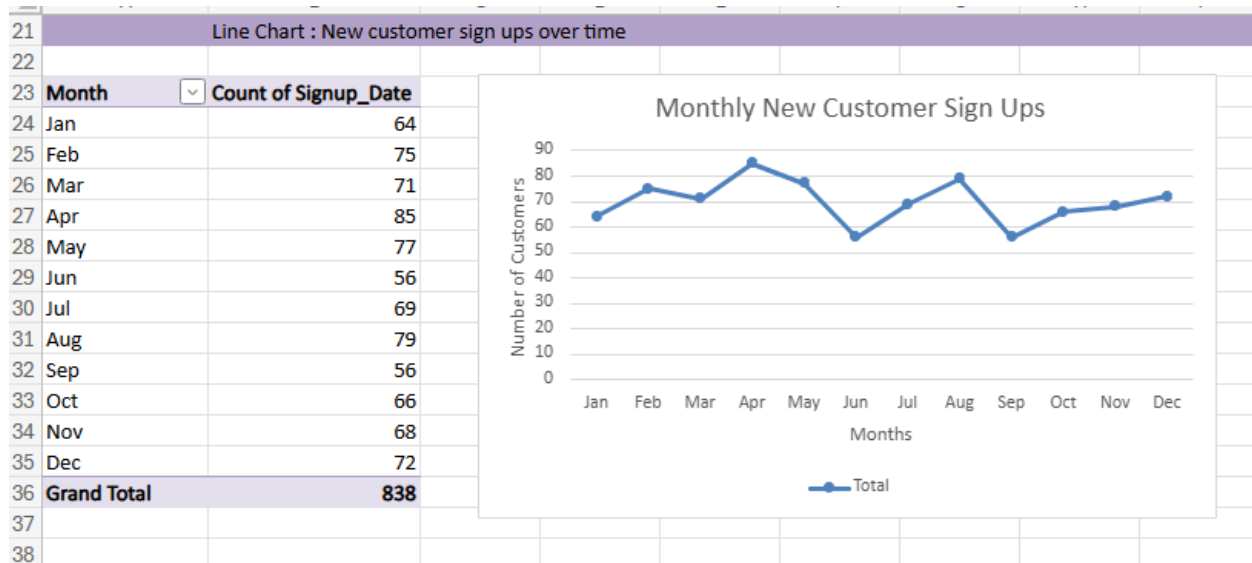
Description

The horizontal bar chart compares total customer spending across different countries.

Analysis

- The **United States** generated the highest total customer spending (**\$50,454.41**), significantly outperforming all other countries.
- The **United Kingdom** ranks second with **\$41,446.38**, highlighting strong revenue contribution from this market.
- The **UAE (\$22,685.98)** and **China (\$21,161.27)** represent the next highest spending countries.

- Countries such as **Egypt, South Korea, Mexico, Germany, and Italy** contribute moderate spending levels ranging from approximately **\$13,000 to \$18,000**.
- **France (\$8,958.56)** recorded the lowest total spend among the analyzed countries.
- The spending pattern largely aligns with customer distribution, where countries with larger customer bases generally generate higher revenue. However, some markets, such as the UAE, demonstrate relatively high spending despite having fewer customers than the United States and United Kingdom, suggesting stronger average spending per customer.



2. Monthly New Customer Sign-Ups

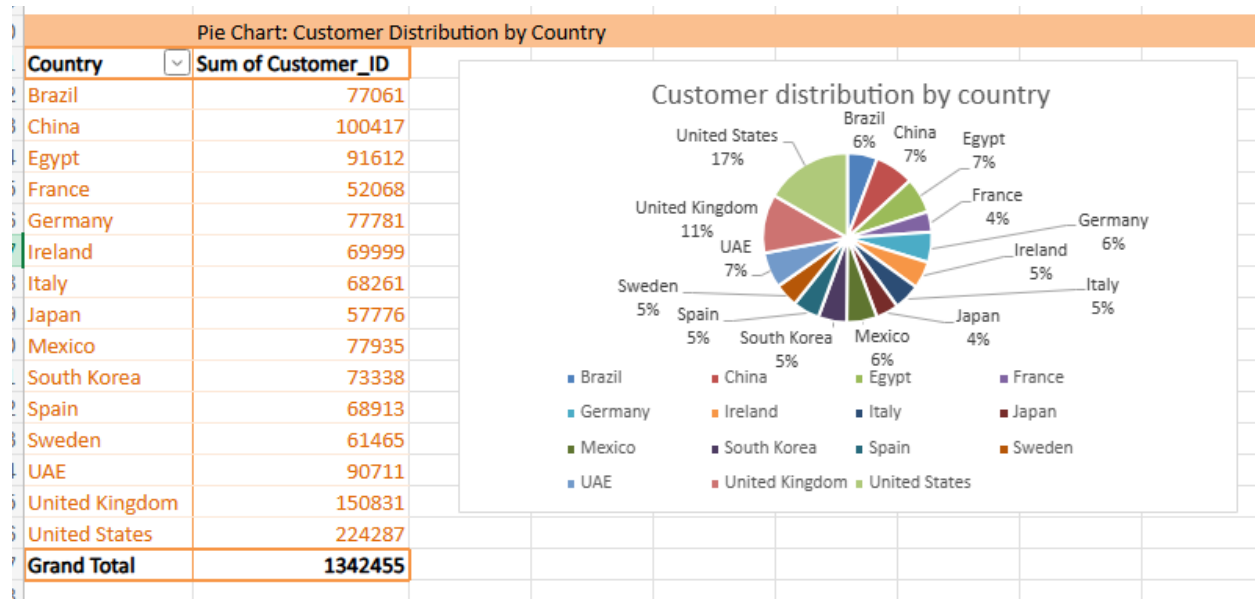
Description

The line chart shows the number of new customer registrations recorded each month throughout the year.

Analysis

- A total of **838 new customers** signed up during the year.
- Customer acquisition peaked in **April (85 sign-ups)**, indicating the strongest month for attracting new customers.
- Other high-performing months include **August (79)** and **May (77)**.
- The lowest sign-up levels occurred in **June (56)** and **September (56)**, suggesting periods of reduced customer acquisition activity.
- Overall, monthly sign-ups remained relatively stable, fluctuating between **56 and 85 customers**, with no extreme increases or declines.

- The recovery in sign-ups after June and September indicates resilience in customer acquisition efforts and consistent market demand.



3. Customer Distribution by Country

Description

The pie chart illustrates the distribution of customers across 13 countries based on the total number of registered customers. The United States has the largest customer base, followed by the United Kingdom and China.

Analysis

- The **United States** accounts for the highest share of customers (**224,287 customers; 17%**), indicating it is the company's largest market.
- The **United Kingdom** contributes **11%** of the total customer base, making it the second-largest market.
- **China (7%), Egypt (7%), Germany (6%), Brazil (6%), and Mexico (6%)** also represent significant customer segments.
- Other countries such as **France, Ireland, Italy, South Korea, Spain, and Sweden** each contribute between **4% and 5%** of total customers.

- The relatively balanced distribution across multiple countries suggests a diversified international customer base, reducing dependence on a single market

Recommendations

- Prioritize customer retention efforts in the United States and United Kingdom.
- Develop targeted marketing campaigns for the 36–45 age segment.
- Investigate customers with low or zero spending activity.
- Implement data validation processes to reduce future data quality issues.

Conclusion

The project successfully transformed raw customer data into a clean and analysis-ready dataset. Through data cleaning, validation, and exploratory analysis, valuable insights were generated regarding customer demographics and spending behavior. The results demonstrate the importance of data quality in supporting reliable business intelligence and informed decision-making.